

Health, Wellness and Sciences Centre

Honours Bachelor of Science – Nursing

Course Information

Course Title	Anatomy and Physiology 1
Course Code	NURS 1016
Course Hours	42

Faculty Information

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Georgian College Land Acknowledgement

Georgian College acknowledges that all campuses are situated on the traditional land of the Anishnaabeg people. The Anishnaabeg include the Odawa, Ojibwe, and Pottawatomi nations, collectively known as the Three Fires Confederacy. Georgian College is dedicated to honouring Indigenous history and culture and committed to moving forward in the spirit of reconciliation and respect with all First Nation, Metis and Inuit people.

These course materials are designed for use as part of the Honours BScN Program at Georgian College and are the property of the instructor unless otherwise stated. Third party copyright materials (such as book chapters, journal articles, music, videos) have either been licensed for use in this course or fall under exception or limitation in Canadian Copyright law. Copying this material for distribution (e.g., uploading material to a commercial third-party website) may lead to violation of copyright law

Course Description:

Students focus on the study of anatomy (structure) and physiology (function) of the human body. The relationship between body organ structures and functions is presented with particular emphasis on homeostasis, cellular metabolism, the organization of body systems, and normal organ functioning. There is an emphasis on applying knowledge of anatomy and physiology to the understanding of health and illness.

Format: In-Class Theory**Core Content:**

- medical and nursing terminology in the context of healthcare
- homeostasis and the principle of complementarity
- biochemical and cellular basis of life
- introduction to human anatomical structures and related terminology
- introduction to human physiology, including individual variations from the norm
- major organ system structure and functioning (e.g., integumentary, skeletal, muscular, nervous, autonomic, and endocrine systems as well as the special senses)

Course Learning Objectives:

Upon successful completion of this course, the student has reliably demonstrated the ability to:

1. locate, identify, and describe the relevant anatomical structures of each organ system;
2. describe and explain the physiological function of each organ system, including important internal and external requirements to sustain normal function;
3. recognize and explain how biochemical, cellular, and homeostatic processes contribute to normal organ functioning;
4. use appropriate medical and nursing terminology relevant to anatomy and physiology in the context of healthcare.

Learning Objective	College of Nurses Entry to Practice Competency
Locate, identify, and describe the relevant anatomical structures of each organ system.	1.21 – T, A, E
Describe and explain the physiological function of each organ system, including important internal and external requirements to sustain normal function.	1.21 – T, A, E
Recognize and explain how biochemical, cellular, and homeostatic processes contribute to normal organ functioning.	1.21 – T, A, E
Use appropriate medical and nursing terminology relevant to anatomy and physiology in the context of healthcare.	1.21 – T, A, E

Prerequisite(s): None

Concurrent Prerequisite(s): NURS 1017 (Pathophysiology 1)

Required Text:

Betts, J. G., Desaix, P., Johnson, E., Johnson, J. E., Korol, O., Kruse, D., Poe, B., Wise, J. A., Womble, M. & Young, K. A. (2022) *Anatomy and Physiology* (2nd ed.). OpenStax.
<https://openstax.org/details/books/anatomy-and-physiology-2e>

Tests and Assignments

Test/Assignment	Percentage of final grade	Due Date	Learning Objective	CNO ETP Competency
Assignment 1: Students will identify various tissue photomicrographs and images of bones as an <u>online assignment</u> . Prior to the activity, the students will utilize Zygot Body to visualize and manipulate the human skeleton.	5	October 12, 2025, at 2359 (Online assignment on Blackboard)	1 and 4	1.21
Assignment 2: Students will view several videos of physiological experiments related to the course content, obtain the relevant data, and write a detailed assignment. The experiments will allow the students to learn core physiological principles. The students will interpret the lab data and formulate answers to the critical-thinking questions.	10	November 2, 2025, at 2359 (Online submission on Blackboard)	2, 3, and 4	1.21
Case Study Assignment: A case study will be presented based on several key physiological concepts in the course. Critical thinking questions based on the case study will be asked. Students will prepare an assignment	10	November 30, 2025, at 2359 (Assignment submission on Blackboard)	2, 3, and 4	1.21

that is based on discussing the posed questions utilizing the knowledge and understanding gained from the course as well as exploring external peer-reviewed scientific resources.				
Quizzes: The online Blackboard quizzes will enable the students to consolidate their learning. The focus of the quizzes will be on the learning outcomes related to the weekly course content. Students will have an opportunity to identify areas of improvement through these formative assessments and remediate them through further practice.	10	Quiz 1 (Chapters 1 – 4): September 21, 2025, at 2359 Quiz 2 (Chapters 5 – 8): October 5, 2025, at 2359 Quiz 3 (Chapters 9 – 11): November 9, 2025, at 2359 Quiz 4 (Chapters 12 and 13): November 23, 2025, at 2359 Quiz 5 (14, 15, and 17): December 7, 2025, at 2359	1, 2, 3, and 4	1.21
Midterm Test: The mid-term test will assess the learning outcomes through the half-way point of the course. The format of the midterm will be a mix of NCLEX-style questions, regular MCQ, select-all-that-apply (SATA) questions, short answer questions, and an essay-type question. Students will obtain constructive feedback from the test to	30	October 14, 2025, in the classroom during class time (2 h long) (Chapters 1 – 10)	1, 2, 3, and 4	1.21

reinforce their learning and develop more effective learning habits. The test also serves as a checkpoint for the students to self-evaluate their progress in the course and utilize the learning resources that are available at the college (e.g., student success services, writing centre, etc.).				
Final Exam: The comprehensive final exam will serve as a summative assessment that evaluates the achievement of the learning outcomes. The students will be provided with practice questions to self-evaluate and devise their own learning plans to achieve the learning outcomes. The format of the final exam will be a mix of NCLEX-style questions, regular MCQ, and select-all-that-apply (SATA) questions.	35	Week 14 (Chapters 1 – 15, 17)		
Total: 100%				

Test and Assignment Descriptions

Please review the Assignments thoroughly so you understand the expectations for each assignment. Please bring any questions about assignments during class or make an individual appointment. Policies and procedure for assignments and exams can be found in your BScN Student handbook or online at [11. Tests and examinations < Georgian College Catalogue](#)

1. Title: Quizzes (online tests on Blackboard)

Due Dates:

Quiz 1 (Chapters 1 – 4); Due Date: September 21, 2025, at 2359; 15 MCQ; 15 minutes

Quiz 2 (Chapters 5 – 8); Due Date: October 5, 2025, at 2359; 15 MCQ; 15 minutes

Quiz 3 (Chapters 9 – 11); Due Date: November 9, 2025, at 2359; 15 MCQ; 15 minutes

Quiz 4 (Chapters 12 and 13); Due Date: November 23, 2025, at 2359; 15 MCQ; 15 minutes

Quiz 5 (Chapters 14, 15, and 17); Due Date: December 7, 2025, at 2359; 15 MCQ; 15 minutes

Description: The online Blackboard quizzes will enable the students to consolidate their learning. The focus of the quizzes will be on the learning outcomes related to the weekly course content. Students will have an opportunity to identify areas of improvement through these formative assessments and remediate them through further practice.

Instructions: Please click on the “Quizzes” folder on Blackboard for further instructions. Each quiz will be a separate online link that you should click on to access. The format of the quiz is 15 multiple choice questions (MCQ) and there is a 15-minute time limit. The questions will be in NCLEX format. One question will be displayed at a time and backtracking is NOT permitted.

2. Title: Assignment 1 (on Blackboard)

Due Date: October 12, 2025, at 2359 (on Blackboard)

Description: Students will identify various tissue photomicrographs and photos of bones as an online assignment. Prior to the activity, the students will utilize Zygote Body to visualize and manipulate the human skeleton.

Instructions: Students will identify various tissues and bones on Blackboard. The assignment consists of 5 tissue photomicrographs and 5 individual bones that must be identified. More instructions will follow prior to the assignment.

3. Title: Assignment 2

Due Date: November 2, 2025, at 2359 (online submission on Blackboard)

Description: Students will view several videos of physiological experiments related to the course content, obtain the relevant data, and write a detailed assignment. The experiments will allow the students to learn core physiological principles. The students will interpret the lab data and formulate answers to the critical-thinking questions.

Instructions: Students will view the physiological experiments and gather and record the relevant data. Students will then answer several questions that are posed in the lab assignment related to the conducted experiments. Students will submit their answers in the form of an online assignment (in APA format) through the provided link. Further instructions will be provided in class as well as on Blackboard.

4. Title: Case Study Assignment

Due Date: November 30, 2025, at 2359 (online submission Blackboard)

Description: A case study will be presented based on several key physiological concepts in the course. Critical thinking questions based on the case study will be asked. Students will prepare an assignment that is based on discussing the questions posed utilizing the knowledge and understanding gained from the course as well as exploring external peer-reviewed scientific resources.

Instructions: Students will submit the case study assignment (in APA format) online through the link provided on Blackboard. Further instructions will be provided in class as well as on the Blackboard.

If extenuating circumstances arise, and you are unable to meet the due date and time for an assignment, you must contact the course professor **prior** to that date/time. If you provide appropriate notice and/or a valid reason in advance of the due date, an extension **may** be negotiated between the student and the professor.

Any late assignments and will be subject to the policies in the BScN Student Handbook

Additional regulations for assignments can be found in your BScN Student Handbook

It is important that you keep any notes and rough work for any papers until the final mark for the course has been recorded. Please be sure to save electronic work frequently while working on an assignment.

Assignments are to be completed independently; this means by one student only. If you choose to work with a peer, and parts of your assignment are similar to the student with whom you worked, you will be risking plagiarism.

5. Title: Midterm test

Due Date: October 14, 2025, during class time in the assigned classroom

Description: The midterm test will assess the learning outcomes from Chapters 1 – 10 inclusive. The length of the midterm will be 2 h long. The format of the midterm is as follows: 32 MCQ, 8 Select-all-that-apply (SATA) questions, short-answer questions (including labelling anatomical diagrams) (30 marks), and 1 physiology-related essay format question (10 marks). The midterm test will be out of 80 marks and will be worth 30% of the overall course grade.

Instructions: The paper-based midterm test will be written in class during class time. The midterm will be 2 hours long. The answers to the MCQ and the SATA questions should be marked on the test booklet as well as on the Akindi sheet. All other answers should be written in the test booklet in the indicated space.

6. Title: Comprehensive Final Exam

Due Date: Week 14

Description: The comprehensive final exam will assess the learning outcomes from the entire course. However, the emphasis of the final exam will be on the content that you have not yet been assessed on (i.e., Chapters 11 – 15, and 17 inclusive). The length of the final exam will be 2 h long. The format of the comprehensive final exam is as follows: 80 MCQ and 10 Select-all-that-apply (SATA) questions. The comprehensive final exam will be out of 90 marks and will be worth 35% of the overall course grade.

Instructions: The paper-based comprehensive final exam will be written during Week 14 in a classroom. The final exam schedule will be posted on Blackboard. The answers to the MCQ and the SATA questions should be marked on the test booklet as well as on the Akindi sheet.

Weekly Flow

Week	Content	Readings Textbook: OpenStax Anatomy and Physiology
1	Introduction to the course Introduction to the Human Body - Anatomical, Medical, and Nursing Terminology - Structural organization of the body - Functions and requirements of life - Organ system overview - Homeostasis The Chemical Level of Organization - Inorganic components (electrolytes, water) - Organic compounds (biomolecules) - Enzymes - Acids and bases	Syllabus Chapter 1 Chapter 2
2	The Cellular Level of Organization - Plasma membrane, organelles - Resting membrane potential - Transport processes The Tissue Level of Organization - Epithelial, connective, and muscle tissue - Classification of the tissues	Chapter 3 Chapter 4
3	The Integumentary System - Layers of the skin - Accessory structures - Physiology functions Assessment: Quiz 1 (Due Date: September 21, 2025, at 2359)	Chapter 5 Chapters 1 - 4
4	Skeletal System – Part 1 Bone structure and classification	Chapter 6

	- Bone growth and ossification Skeletal System – Part 2 - Axial skeleton - Appendicular skeleton	Chapter 7 Chapter 8
5	Joints - Structural and functional classification - Fibrous and cartilaginous joints - Synovial joints (focus on the knee joint) - Types of body movement Assessment: Quiz 2 (Due Date: October 5, 2025, at 2359)	Chapter 9 Chapters 5 - 8
6	Muscle Physiology - Types of muscle fibers - Skeletal muscle contraction (events at neuromuscular junction, excitation-contraction coupling, and cross-bridge cycling) - Muscle fatigue Assessment: Assignment 1 (Due Date: October 12, 2025, at 2359)	Chapter 10 Chapters 4, 7, and 8
7	Assessment: Midterm test (October 14, 2025)	Chapters 1 – 10
Study week		
8	Muscular System - Axial muscles of the head, neck, and back - Axial muscles of the abdominal wall and thorax - Appendicular muscles of the pectoral girdle and arms - Appendicular muscles of the pelvic girdle and legs Assessment: Assignment 2 (Due Date: November 2, 2025, at 2359)	Chapter 11 Assignment 2 (on Blackboard)

9	The Nervous System • Types of neurons and neuroglial cells • Generation of an action potential • Synapses and neurotransmitters Central Nervous System • Brain (cortical and subcortical structures) • Meningeal layers • Ventricular system and CSF flow • Cranial nerves and their functions (important) Assessment: Quiz 3 (Due Date: November 9, 2025, at 2359)	Chapter 12 Chapters 13 Chapters 9 – 11
10	Peripheral Nervous System • Spinal cord structure • Spinal nerves and nerve plexuses (cervical, brachial, lumbar, and sacral)	Chapter 13
11	Somatic Nervous System • Sensory pathways • Motor pathways • Special senses (vision, hearing, and balance/equilibrium) Assessment: Quiz 4 (Due Date: November 23, 2025, at 2359)	Chapter 14 Chapters 12 and 13
12	Autonomic Nervous System • Structure and function of the sympathetic nervous system • Structure and function of the parasympathetic nervous system Endocrine system • Hypothalamus and pituitary gland and their related hormones • Thyroid gland and associated hormones • Adrenal cortex and associated hormones • Parathyroid gland and associated hormones	Chapter 15 Chapter 17

	Pancreas and associated endocrine hormones Assessment: Case Study Assignment (Due Date: November 30, 2025, at 2359)	Case Study Assignment (on Blackboard)
13	Assessment: Quiz 5 (Due Date: December 7, 2025, at 2359)	Chapters 14, 15, and 17
14	Assessment: Comprehensive Final Exam	All chapters listed above but the emphasis will be on Chapters 11 – 15, 17

*Due to unforeseen circumstances, this schedule may be adjusted as needed by faculty. Any changes will be communicated to all students in class and/or on Blackboard

In-person Class Cancellation Protocol

If the in-person class is cancelled by the college due to the weather (or other events) (see Syllabus addendum on p. 27), I will make every effort to hold a virtual class online instead. If the in-person class is cancelled by myself due to illness, I will also make every effort to conduct a virtual class online instead. Instructions for attending the virtual class will be emailed to the students and posted as an announcement on Blackboard.

Professionalism and Academic Integrity

Academic integrity is a core value at Georgian College and a priority in all academic activities. Students are responsible for being aware of and demonstrating behaviour that is honest and ethical. Students are also expected to take responsibility for their own academic work, adhering to integrity standards for themselves and their program but also encouraging and cultivating a culture of integrity among their classmates.

There are many significant offenses that warrant academic misconduct. Plagiarism and cheating for instance are behaviors that pose strong concern for student success. For further elaboration on behaviors of Academic Misconduct, refer to the Academic Calendar, Section 8.0 on [Academic Misconduct](#)).

Expectations

The BScN program embraces a philosophy of equity, diversity, and inclusiveness in all interactions. A positive learning environment is integral to learning. Please refer to the [Student Code of Conduct](#) and program handbook for further information.

Student Assistance and Accommodations

If you require further assistance with your learning, please discuss it with your faculty, and/or seek assistance from Student Services. Information about Student Services and useful resources will be posted on your Semester 1 Blackboard organization and is also available on the College website. A good person to start with is your Student Advisor

<http://www.georgiancollege.ca/student-life/student-services/student-advising/>

The student advisor for nursing is Guisselle Mejia Lobo (Guisselle.MejiaLobo@GeorgianCollege.ca). Please contact the advisor if you need assistance with your learning that is not directly related to the course content (e.g., study habits, time management, etc.)

If personal difficulties such as depression, anxiety, or other problems are affecting your ability to learn, please make an appointment with a counselor in Student Services

<https://www.georgiancollege.ca/student-life/student-services/counselling/>. These services are free and completely confidential and can assist you to be a successful student.

Please do not wait until it is too late in the semester to seek assistance. The sooner you can make use of available resources, the better your chance of success. There is no reason not to consider the many services offered to you at Georgian College.

Maximizing your Grade

Make use of the College resources to help you become a better student. Georgian College has a wide variety of resources and support staff to help you become a more effective student. If you experience any difficulty with studying, writing tests or learning course material, consider making an appointment - [Contact the Library - Academic Success - Library and Academic Success at Georgian College](#)

Chapter Learning Outcomes

Chapter 1 (An Introduction to the Human Body) Learning Outcomes

1. Utilize appropriate medical and nursing terminology related to the structure and function of human body systems.
2. Define anatomy and physiology and describe their subdivisions.
3. Explain the principle of complementarity.
4. Name the different levels of structural organization that make up the human body and explain their relationships.
5. List the functional characteristics necessary to maintain life in humans.
6. List the survival needs of the body.
7. List the eleven (11) organ systems of the body, identify their components, and briefly explain the major function(s) of each system.
8. Define homeostasis and explain its significance.
9. Identify and describe the necessary components of a homeostatic feedback loop.
10. Describe how negative and positive feedback maintain homeostasis.
11. Describe the anatomical position.
12. Use correct anatomical terms to describe body directions, regions, and body planes or sections.
13. Locate and name the major body cavities and their subdivisions.
14. Name the four quadrants of the abdominopelvic cavity and list the organs they contain.

Required Readings: Chapter 1

Chapter 2 (The Chemical Level of Organization) Learning Outcomes

1. Explain the importance of water and electrolytes in homeostasis.
2. Define an acid and a base, and explain the concept of pH.
3. Explain the importance of acid-base homeostasis.
4. Describe and compare the building blocks, general structures, and biological functions of carbohydrates, proteins, lipids, and nucleic acids.
5. Describe the four levels of protein structure.
6. Define an enzyme and describe enzyme action.
7. Explain the role of ATP in cell metabolism.

Required Readings: Chapter 2

Chapter 3 (The Cellular Level of Organization) Learning Outcomes

1. Utilize appropriate medical and nursing terminology related to the structure and function of the cell.
2. Define the cell.
3. Describe the composition of the plasma membrane and relate it to membrane functions.
4. Compare the structure and function of tight junctions and gap junctions.
5. Relate the plasma membrane to active and passive transport processes.
6. Compare and contrast simple diffusion, facilitated diffusion, and osmosis relative to substances transported, direction, and mechanism.
7. Differentiate between primary and secondary active transport.
8. Describe the composition of the cytosol.
9. Discuss the function of the mitochondria, ribosomes, the rough and smooth endoplasmic reticulum, the Golgi apparatus, and lysosomes.
10. Outline the structure and function of the nuclear envelope, nucleolus, and chromatin.
11. Very briefly describe the transcription of DNA into mRNA and the translation of mRNA into proteins.

Required Readings: Chapter 3

Chapter 4 (The Tissue Level of Organization) Learning Outcomes

1. List the structural and functional characteristics of epithelial tissue.
2. Name, classify, and describe the various types of epithelia, and briefly indicate their chief function(s) and location(s).
3. Describe the different types of connective tissue and their typical locations in the body.
4. Identify the structural elements and characteristics functions of the different connective tissues.
5. Compare and contrast the structures and body locations of skeletal, cardiac, and smooth muscle tissues.

Required Readings: Chapter 4

Chapter 5 (The Integumentary System) Learning Outcomes

1. Utilize appropriate medical and nursing terminology related to the structure and function of the integumentary system.
2. Name the tissue types composing the epidermis and dermis.
3. List and describe the properties and functions of the epidermis and dermis.
4. Describe the factors that contribute to skin colour.
5. Identify the various touch and pain receptors.
6. Identify the locations of eccrine and apocrine sweat glands as well as the sebaceous oil glands.
7. Describe the physiological functions of the integumentary system.

Required Readings: Chapter 5

Chapter 6 (Bone Tissue and the Skeletal System) Learning Outcomes

1. Utilize appropriate medical and nursing terminology related to the structure and function of the skeletal system.
2. Describe the functional properties of the three types of cartilage tissue.
3. Locate the major cartilages of the adult skeleton.
4. Name the major regions of the skeleton and describe their relative functions.
5. Compare and contrast the four bone classes and provide examples of each class.
6. Describe the gross anatomy of a typical long bone.
7. Describe the histology of compact bone.
8. Discuss the chemical composition of bone.
9. Describe the functions of the three types of bone cells.
10. Discuss the process of endochondral ossification.
11. Describe the process of long bone growth that occurs at the epiphyseal plates.
12. Describe the anatomical and physiological functions of bones.

Required Readings: Chapter 6

Chapters 7 and 8 (The Skeletal System) Learning Outcomes

1. Utilize appropriate medical and nursing terminology related to the structure and function of the skeletal system.
2. Name the major parts of the axial and appendicular skeletons and describe their relative functions.
3. Identify, locate, and describe the cranial and facial bones.
4. Identify, locate, and describe the sutures of the skull.
5. Describe the structure of the vertebral column, list its components, and describe its curvatures.
6. Discuss the structure of a typical vertebra and describe regional features of cervical, thoracic, and lumbar vertebrae.
7. Identify, locate, and describe the bones of the thoracic cage.
8. Identify, locate, and describe the bones that form the pectoral girdle and relate their structure and arrangement to the function of this girdle.
9. Identify, locate, and describe the bones of the upper limb.
10. Name the bones contributing to the os coxae and relate the pelvic girdle's strength to its function.
11. Describe differences in the male and female pelvis and relate these to functional differences.
12. Identify, locate, and describe the bones of the lower limb.

Required Readings: Chapters 7 and 8

Chapter 9 (Joints) Learning Outcomes

1. Utilize appropriate medical and nursing terminology related to the structure and function of joints.
2. Define the terms: (i) joint; (ii) articulation.
3. Classify joints by structure and function.
4. Describe the general structure of fibrous joints.
5. Name and give an example of each of the three common types of fibrous joints.
6. Describe the general structure of cartilaginous joints.
7. Name and give an example of each of the two common types of cartilaginous joints.
8. Describe the structural characteristics of synovial joints.
9. Name and describe (and perform) the common body movements.
10. Name and provide examples of the six types of synovial joints based on the movement(s) allowed.
11. Describe, in detail, the knee joints in terms of articulating bones, ligaments, and other anatomical characteristics of the joint.

Required Readings: Chapter 9

Chapter 10 (Muscle Tissue) Learning Outcomes

1. Utilize appropriate medical and nursing terminology related to the structure and function of the muscular system.
2. Compare and contrast the three basic types of muscle tissue. (Note: Cardiac muscle will be covered in NURS 1020)
3. List the important functions of muscle tissue.
4. Describe the gross structure of a skeletal muscle.
5. Describe the microscopic structure and functional roles of the myofibrils, sarcoplasmic reticulum, and T tubules of skeletal muscle fibers.
6. Describe the sliding filament model of muscle contraction.
7. Explain how muscle fibers are stimulated to contract by describing events that occur at the neuromuscular junction.
8. Describe how an action potential in the muscle is generated.
9. Follow the events of excitation-contraction coupling that lead to cross bridge activity.
10. Describe three ways in which ATP is regenerated during skeletal muscle contraction.
11. Describe the probable causes of muscle fatigue.
12. Briefly describe three types of skeletal muscle fibers and explain the function of each type.

Required Readings: Chapter 10

Chapter 11 (The Muscular System) Learning Outcomes

1. Utilize appropriate medical and nursing terminology related to the structure and function of the muscular system.
2. Briefly describe the functions of prime movers and antagonists.
3. Briefly explain how a muscle's position relative to a joint affects its action.
4. Identify, locate, and name specific muscles in anatomical diagrams. The supplementary handout lists the muscles that you must learn.
5. Briefly state the action(s) of each muscle in the supplementary handout.

Required Readings: Chapter 11

Chapter 12 (The Nervous System and Nervous Tissue) Learning Outcomes

1. Utilize appropriate medical and nursing terminology related to the structure and function of the nervous system.
2. Describe the basic functions of the nervous system.
3. Explain the structural and functional divisions of the nervous system.
4. List the types of neuroglial cells and describe their functions.
5. Define a neuron, describe its important structural components, and relate each part of a neuron to a functional role.
6. Explain the importance of the myelin sheath and describe how it is formed in the central and peripheral nervous systems.
7. Classify neurons structurally and functionally.
8. Define the resting membrane potential and describe its electrochemical basis.
9. Explain how action potentials are generated and propagated along neurons.
10. Define absolute and relative refractory periods.
11. Define saltatory conduction and contrast it to conduction along unmyelinated fibers.
12. Define a synapse. Explain the structure and physiology of a chemical synapse.
13. Distinguish between excitatory and inhibitory neurons.
14. Name several important excitatory and inhibitory neurotransmitters.

Required Readings: Chapter 12

Chapter 13 (Anatomy of the Nervous System) Learning Outcomes

Chapter 14 (The Somatic Nervous System) Learning Outcomes

1. Utilize appropriate medical and nursing terminology related to the structure and function of the nervous system and the special senses.
2. Identify and describe the structure and function (briefly) of the major regions of the adult brain.
3. Name and locate the ventricles of the brain.
4. List the major lobes, fissures, and functional areas of the cerebral cortex.
5. Describe the general function of the basal nuclei (basal ganglia).
6. Describe the location of the diencephalon and name its subdivisions and functions.
7. Identify the three major regions of the brain stem and describe the functions of each area.
8. Describe the structure and function of the cerebellum.
9. Locate the limbic system and the reticular formation and explain the role of each functional system.
10. Identify and describe the layers of the meninges.
11. Describe how the meninges, cerebrospinal fluid, and the blood brain barrier protect the CNS.
12. Explain how cerebrospinal fluid is formed and describe its circulation through the brain and spinal cord.
13. Describe the gross and microscopic structure of the spinal cord.
14. Briefly list a few major spinal cord tracts and classify each as a motor or sensory tract.
15. Define the peripheral nervous system and list its components.
16. Briefly classify general sensory receptors by structure, stimulus detected, and body location.
17. Briefly outline the events that lead to sensation and perception.
18. Describe the general structure of a nerve.
19. Name the 12 pairs of cranial nerves; indicate the body region and structures innervated by each.
20. Describe the function of each of the 12 pairs of cranial nerves.
21. Describe the general structure of a spinal nerve and the general distribution of its rami.
22. Define a plexus. Describe the distribution and function of the peripheral nerves arising from the cervical, brachial, lumbar, and sacral plexus.
23. Describe the anatomy and function of the eye structures, eye layers, the lens, and the humors of the eye.

24. Trace the pathway of light through the eye to the retina and explain how light is focused for distant and close vision.
25. Compare and contrast the roles of rods and cones in vision.
26. Describe the structure and general function of the outer, middle, and inner ear.
27. Explain the functioning of the cochlear and its associated substructures.
28. Explain how the balance organs of the semicircular canals and the vestibule help maintain equilibrium.

Required Readings: Chapters 13 and 14

Chapter 15 (The Autonomic Nervous System) Learning Outcomes

1. Utilize appropriate medical and nursing terminology related to the structure and function of the autonomic nervous system.
2. Define the autonomic nervous system and explain its relationship to the rest of the nervous system.
3. Compare the somatic and autonomic nervous systems relative to effectors, efferent pathways, and neurotransmitters released.
4. Compare and contrast the functions of the parasympathetic and sympathetic divisions of the autonomic nervous system.
5. For the parasympathetic and sympathetic divisions, briefly describe the site of CNS origin, locations of ganglia, and general fiber pathways.
6. Describe the physiological effects of the parasympathetic and sympathetic divisions on the various organs/organ systems.

Required Readings: Chapter 15

Chapter 17 (The Endocrine System) Learning Outcomes

1. Utilize appropriate medical and nursing terminology related to the structure and function of the endocrine system.
2. Indicate important differences between hormonal and neural controls of body functioning.
3. List the major endocrine organs and describe their body locations.
4. Explain how hormone release is regulated.
5. Describe structural and functional relationships between the hormones released by the hypothalamus, the pituitary gland, and the target endocrine organ.
6. Describe the role of negative feedback in the regulation of hormonal release.
7. Briefly describe the function of the two hormones released by the posterior pituitary gland.
8. List and describe the chief effects of the anterior pituitary hormones.
9. Describe the effects of the hormones produced by the thyroid gland.
10. Describe the general functions of parathyroid hormone.
11. List the hormones produced by the adrenal gland and describe their physiological effects.
12. Compare and contrast the effects of the two major pancreatic hormones (insulin and glucagon) related to glucose homeostasis.

Required Readings: Chapter 17

Syllabus Addendum

VPA-Protocol-05

Effective Date: April 1, 2016

Revision Dates: June 3, 2025 **IMPORTANT**

DATES:

Fire and Lockdown drill dates:

- Fall 2024: September 24
- Winter 2025: January 28
- Summer 2025: May 27 **PROCESS:**

Emergency Evacuation

- Evacuate buildings when a fire alarm is activated, or an official announcement is given. Review [emergency evacuation](#).
- ***Students requiring assistance in emergency situations must inform their faculty during the first week of class.***
- Familiarize yourself with all fire exit doors of classrooms and buildings you may occupy.
- Do not re-enter a building until the Fire Department or college personnel give instructions.

Lockdown

- Lockdown is initiated when there is a potential or actual violent incident on campus that could result in a serious injury or threat to life.
- Students may download the safe@georgian app to stay updated on Campus Safety Services' information including lockdown.
- Familiarize yourself with the [College Lockdown Procedure](#).
- Lockdown tests occur each semester.

RESOURCES:

- [Policy/Procedure #57 Unscheduled campus closures](#)
- [Get Out, Hide, Fight Lockdown Video](#)
- Lockdown Model – Get Out, Hide, Fight: [Lockdown Tools and Tactics and FAQs](#) • [Emergency Response Guide](#)

MEETING YOUR RESPONSIBILITIES AS A STUDENT UNDER THE STUDENT CODE OF CONDUCT:

A safe, respectful environment is integral to learning. It is important to always maintain civility for others. Should a student engage in disruptive or other inappropriate behaviours, the matter may be referred for resolution under the Student Code of Conduct Resolution process. Students are encouraged to review their rights, responsibilities and process options as outlined in [the Student Code](#)

of Conduct, as well as Georgian's academic regulation section 8.1.2 concerning their academic responsibilities.

UNSCHEDULED CAMPUS CLOSURE:

Reference: Policy/Procedure #57 Unscheduled campus closures **Timing of Closures/Notification:**

Closure	Decision By	Communication / Notification By	Notes
College has made the decision to close a building, campus or location, or suspend a service operation <u>in the morning</u> :	6:00 a.m.	6:30 a.m.	If re-opening for noon or evening classes is being considered, this will be mentioned in the morning communication
Decisions surrounding <u>evening</u> closures:	2:30 p.m.	3:00 p.m.	
College closes a campus(s) or operation in the morning and <u>expects to re-open by 12:00 noon</u>	9:30 a.m.	10:00 a.m.	Only affects classes beginning at 12:00 noon or later
Closure expected to continue <u>past 12:00 noon</u>	9:30 a.m.	10:00 a.m.	
College intends to <u>re-open for evening classes</u> which commence at 5 p.m. or later	2:30 p.m.	3:00 p.m.	
College intends to <u>NOT re-open for evening classes</u> :	2:30 p.m.	3:00 p.m.	

Notification will be made via:

- Georgian social media (Facebook, Twitter)
- Safe@Georgian app
- Georgian website
- Recorded message when callers telephone a local campus
- Student or employee portal
- Georgian email account
- Staff, students and the public may also hear about closures on local radio. They should monitor more than one communication channel.

COURSE MATERIAL COSTS:

A list of learning material costs related to this course can found [here under Course Material Costs](#). It can also be found on your Banner Booklist.
